

HEPTAPOD B — Complex Plane Parameterization

Step 1 of 5: Preprocessing Pipeline

STEP 1
Original PNG



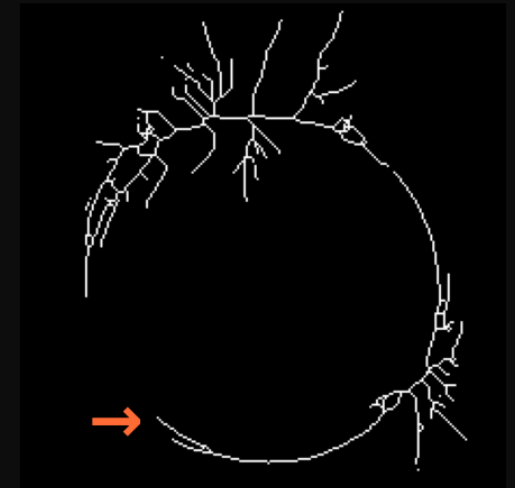
STEP 2
Invert Colors



STEP 3
Binary Threshold



STEP 4
Skeleton / Medial Axis



Raw 224×224 pixel image
Black ink on white bg

Ink becomes white (=1)
Background becomes black (=0)

Hard cutoff at 127
Clean ink mask

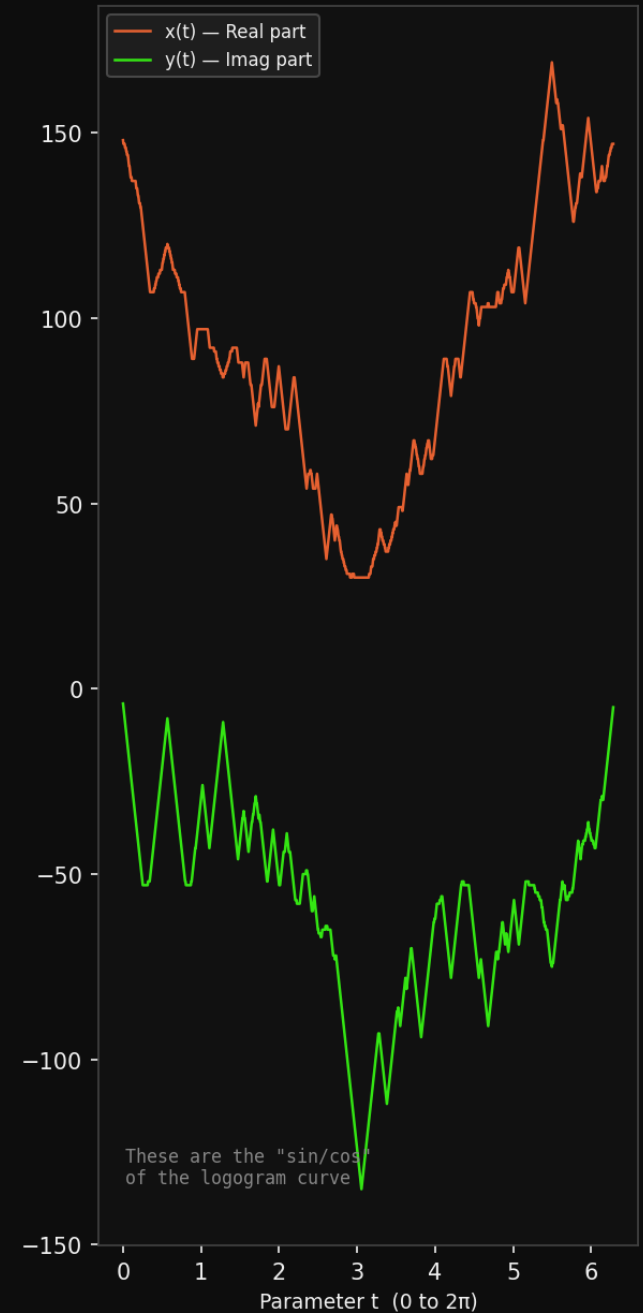
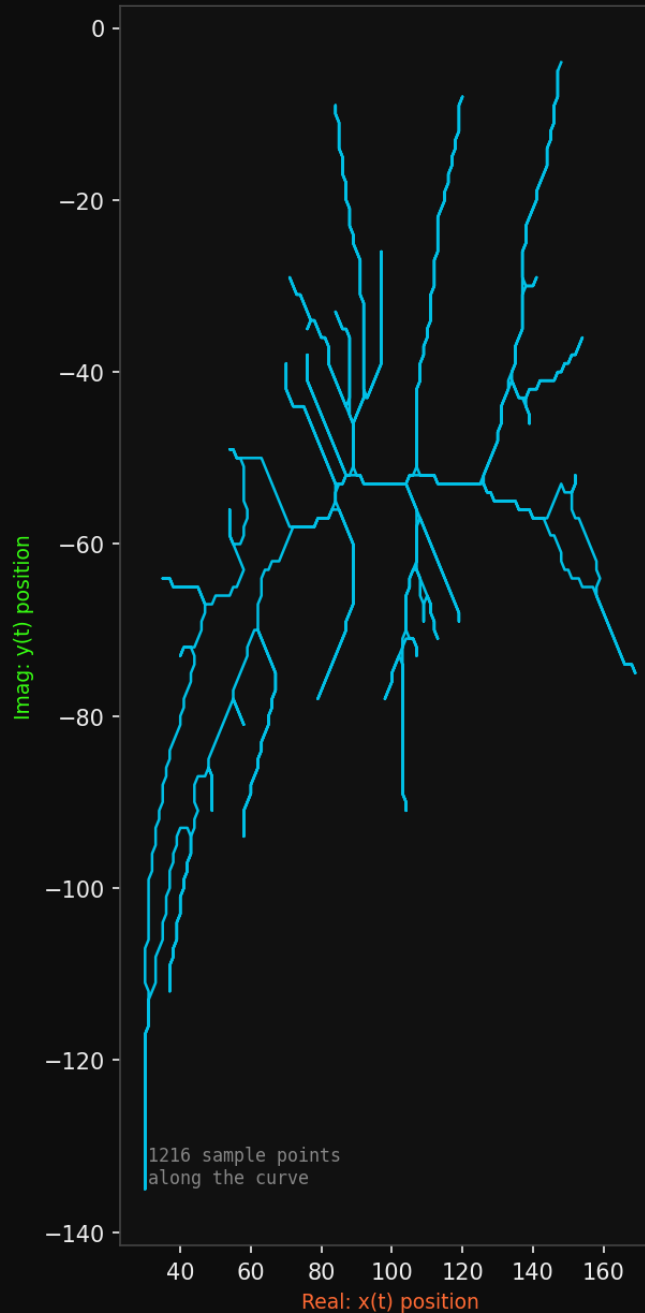
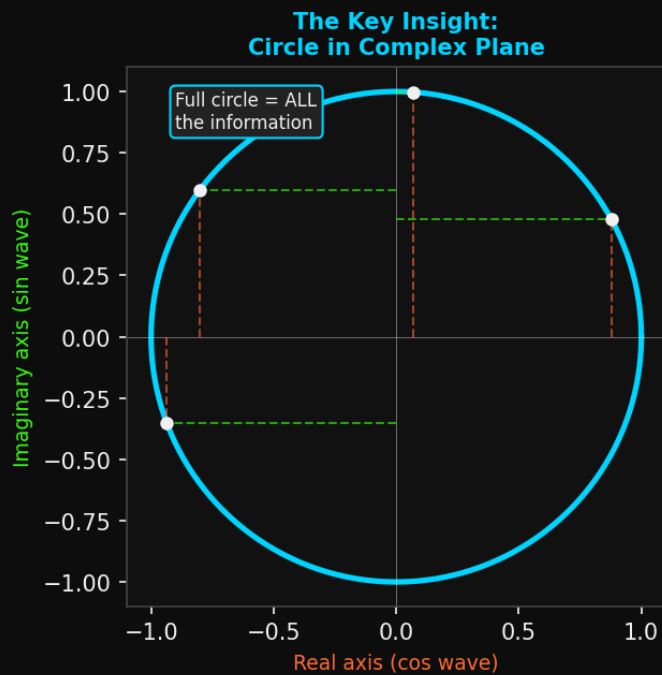
Stroke reduced to 1px centerline
True geometric curve

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Step 2 of 5: The Complex Plane Lift $\gamma(t) = x(t) + i \cdot y(t)$

Logogram Outer Curve
Lifted: $\gamma(t) = x(t) + i \cdot y(t)$

The Two Shadow Signals
(what 2D alone would see)



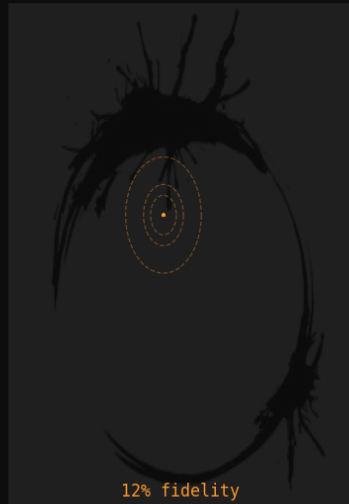
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Step 3 of 5: Fourier Decomposition — Sum of Rotating Circles

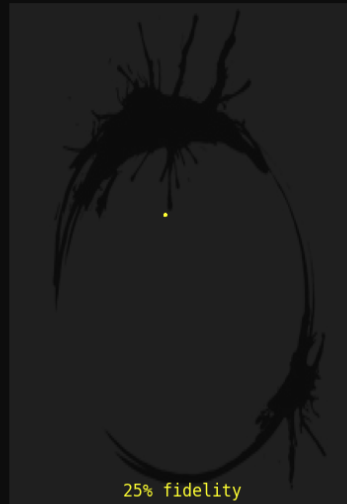
N = 4
9 harmonics



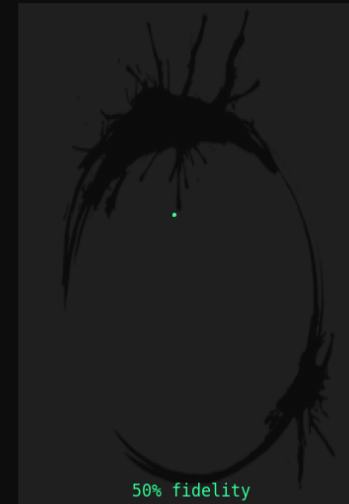
N = 8
17 harmonics



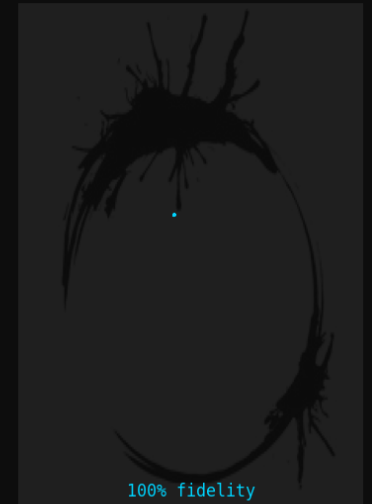
N = 16
33 harmonics



N = 32
65 harmonics

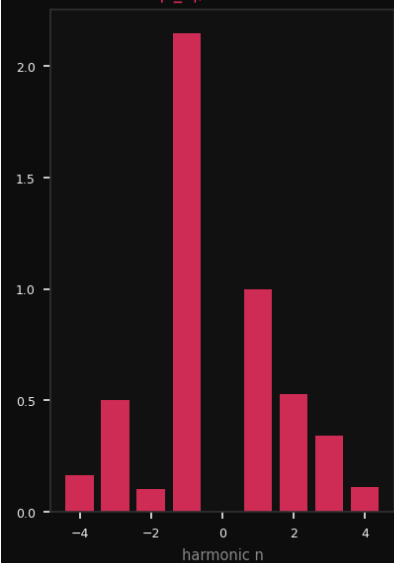


N = 64
129 harmonics

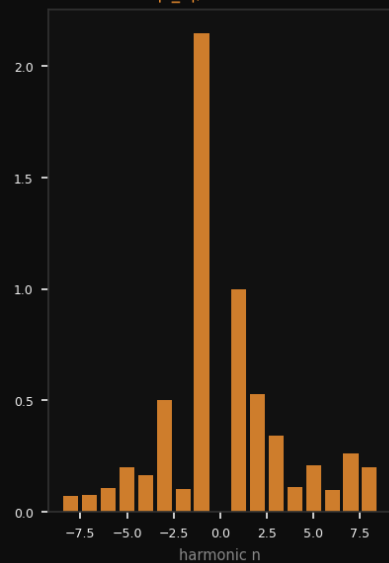


← Low harmonics = basic shape High harmonics = fine detail →

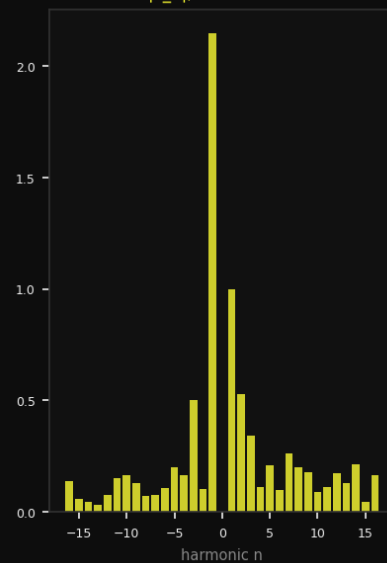
Coefficients used
 $|c_n|$, $n=-4$ to 4



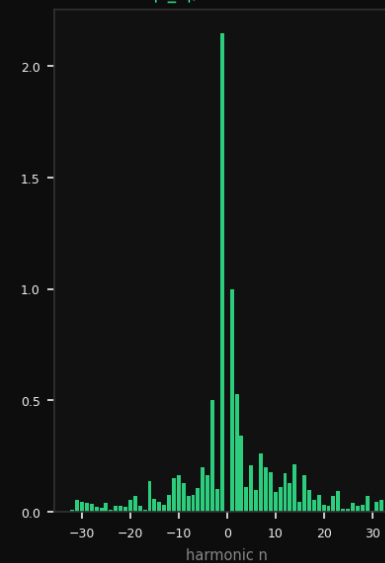
Coefficients used
 $|c_n|$, $n=-8$ to 8



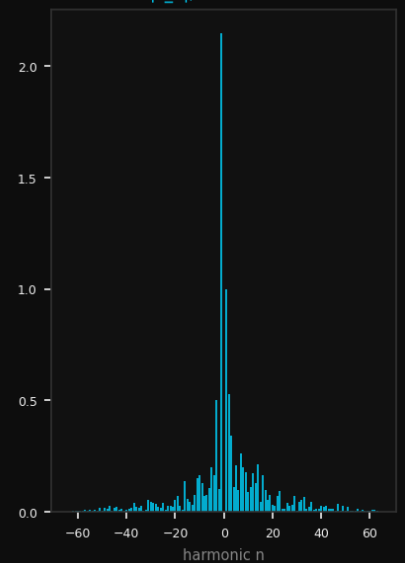
Coefficients used
 $|c_n|$, $n=-16$ to 16



Coefficients used
 $|c_n|$, $n=-32$ to 32



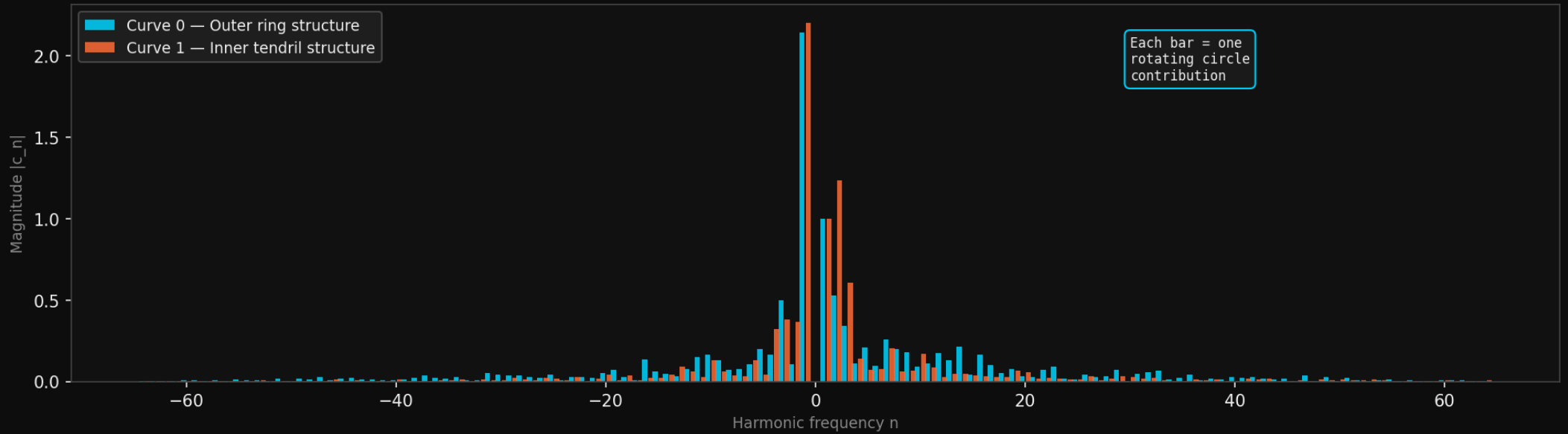
Coefficients used
 $|c_n|$, $n=-64$ to 64



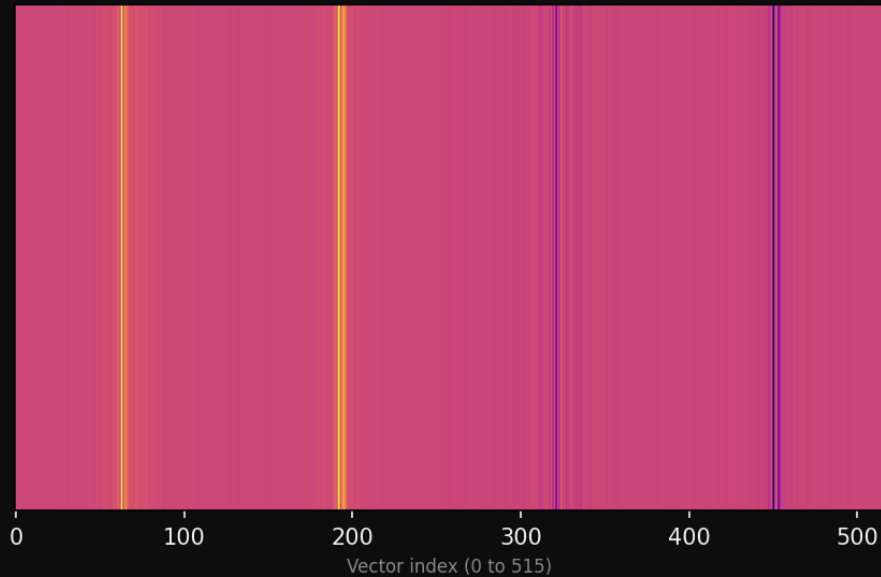
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Step 4 of 5: The Encoding $\Phi(L)$ — Logogram Fingerprint

Full Frequency Fingerprint of Both Curves — This IS the $\Phi(L)$ encoding



$\Phi(L)$ — The Flat Vector
(516-dimensional ML input)



WHAT $\Phi(L)$ ENABLES:

- ✓ Unique fingerprint per logogram
- ✓ Invertible → reconstruct original
- ✓ Same size for all 49 logograms
- ✓ Interpolate between logograms
- ✓ Direct GAN / ML model input

DIMENSIONS:

2 curves × 129 harmonics × 2
= 516 real floats per logogram

49 logograms → 49 × 516 matrix
→ ready for ML pipeline

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Step 5 of 5: Full System Architecture

